

⌈ Preliminary,

Let V be a vector space.

⌈ Def. Suppose $V = W_1 \oplus W_2$, and define a map

$$T: V \rightarrow V; x_1 + x_2 \mapsto x_1 \quad (x_1 \in W_1, x_2 \in W_2)$$

It is called projection along W_2 onto W_1 .

⌈ Lemma. Let $T: V \rightarrow V$ be a projection along W_2 onto W_1 .

1). T is linear operator

2). $W_1 = \text{Im } T = \ker(T - I)$

3). $W_2 = \ker T$.

4). $T^2 = T$.

Proof. Let $x, y \in V$ and $\alpha \in F$ be given. Since $V = W_1 \oplus W_2$,

$$\exists! x_1, y_1 \in W_1, x_2, y_2 \in W_2 \text{ s.t. } x = x_1 + x_2 \text{ and } y = y_1 + y_2.$$

Proof of 1).

$$T(\alpha x + y) = T(\alpha x_1 + \alpha x_2 + y_1 + y_2) = T(\alpha x_1 + y_1 + \alpha x_2 + y_2) = \alpha x_1 + y_1 = \alpha T(x) + T(y).$$

Proof of 2),

$$\begin{aligned} \text{Im } T &= \{T(x_1 + x_2) \mid x_1 \in W_1, x_2 \in W_2\} \\ &= \{T(x_1) \mid x_1 \in W_1\} \\ &= \{T(x) \mid x = T(x)\} \\ &= \{x \mid T(x) = x\} \end{aligned}$$

Proof of 3). Clearly $W_2 \subseteq \ker T$. To show $W_2 \supseteq \ker T$, let $x \in \ker T$ be given.

If $x \notin W_2$, then $x = x_1 + x_2$ for some $x_1 \in W_1, x_2 \in W_2$ s.t. $x_1 \neq 0$.

But, then $T(x) = T(x_1 + x_2) = x_1 = 0$, Contradiction.

Proof of 4). Let $x = x_1 + x_2 \in V$ where $x_1 \in W_1, x_2 \in W_2$ be given.

$$T^2(x) = T(T(x)) = T(x_1) = x_1 = T(x).$$

⌈ In general, for linear map $T: V \rightarrow V$, $\ker T \cap \operatorname{Im} T \neq \{0\}$.

For instance, let

$$T = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \text{ be a Matrix.}$$

Then,

$$\ker T = \left\{ a \begin{pmatrix} 1 \\ 0 \end{pmatrix} \mid a \in F \right\} \text{ and } \operatorname{Im} T = \left\{ b \begin{pmatrix} 1 \\ 0 \end{pmatrix} \mid b \in F \right\}$$

That is, these are same. $\ker T = \operatorname{Im} T$.

⌈ Counterexample for $W_1 \oplus W_2 = W_1 \oplus W_2'$ but $W_2 \neq W_2'$.

Let $V = F^2$, and $W_1 = \langle (1, 0) \rangle_F$.

Put $W_2 = \langle (1, 1) \rangle_F$ and $W_2' = \langle (1, -1) \rangle_F$.

⌈ Lemma. $T: V \rightarrow V$ be a linear on fin. dim V , then

$$V = \ker T + \operatorname{Im} T \text{ or } \ker T \cap \operatorname{Im} T = \{0\} \text{ iff } V = \ker T \oplus \operatorname{Im} T.$$

Lemma. Let V be a vector space, $T: V \rightarrow V$ be a linear operator.

$T^2 = T$ if and only if T is a projection along $\ker T$ onto $\ker(T-I)$.

Proof. Suppose $T^2 = T$. Let $x \in V$ be given. Then,

$$x = x - T(x) + T(x)$$

where $x - T(x) \in \ker T$ and $T(x) \in \ker(T-I)$ because

$$T(x - T(x)) = T(x) - T^2(x) = 0 \quad \text{and} \quad (T-I)(T(x)) = T^2(x) - T(x) = 0.$$

Moreover, let $y \in \ker T \cap \ker(T-I)$. Then

$$0 = T(y) = y$$

Therefore $\ker T \cap \ker(T-I) = \{0\}$, so

$$V = \ker T \oplus \ker(T-I)$$

Let $x_1 \in \ker T$ and $x_2 \in \ker(T-I)$ be given. Then

$$T(x_1 + x_2) = T(x_2) = x_2.$$

Consequently, T is a projection. The converse holds for general projections.

Let $V = W_1 \oplus W_2$, and $T: V \rightarrow V$ be a projection along W_2 onto W_1 . For $x_1 \in W_1, x_2 \in W_2$

$$T(x_1 + x_2) = x_1 = T(x_1) = T^2(x_1) = T^2(x_1 + x_2)$$

Thus $T = T^2$.

Def. Let $(V, \langle \cdot, \cdot \rangle)$ be an inn. prod. space.

A projection $T: V \rightarrow V$ is called orthogonal projection if:

$$(\ker T)^\perp = \text{Im } T \quad \text{and} \quad (\text{Im } T)^\perp = \ker T$$

Thm. Let $T: V \rightarrow V$ be a linear operator on inner product space $(V, \langle \cdot, \cdot \rangle)$.

T is orthogonal projection if and only if adjoint T^* exists, and $T^2 = T = T^*$.

Proof. Since T is projection, iff $T^2 = T$, enough to show that

$$(\ker T)^\perp = \text{Im } T \quad \text{and} \quad (\text{Im } T)^\perp = \ker T \quad \text{iff} \quad T^* \text{ exists and } T = T^*.$$

Let $x, y \in V$ be given. Since $V = \ker T \oplus \text{Im } T$, for some $x_1, y_1 \in \ker T$ and $x_2, y_2 \in \text{Im } T$,

$$x = x_1 + x_2 \quad \text{and} \quad y = y_1 + y_2.$$

Now,

$$\langle x, T(y) \rangle = \langle x_1 + x_2, T(y_1 + y_2) \rangle = \langle x_1 + x_2, y_2 \rangle = \langle x_1, y_2 \rangle + \langle x_2, y_2 \rangle = \langle x_2, y_2 \rangle$$

$$\langle T(x), y \rangle = \langle T(x_1 + x_2), y_1 + y_2 \rangle = \langle x_2, y_1 + y_2 \rangle = \langle x_2, y_1 \rangle + \langle x_2, y_2 \rangle$$

These are same. Therefore, $T = T^*$.

Conversely, let $x \in \ker T$ and $y \in \text{Im } T$ be given. Then,

$$\langle x, y \rangle = \langle x, T(y) \rangle = \langle T^*(x), y \rangle = \langle T(x), y \rangle = \langle 0, y \rangle = 0.$$

$$W \subseteq W^{\perp\perp}$$

Therefore $\ker T \subseteq (\text{Im } T)^\perp$ and $\text{Im } T \subseteq (\ker T)^\perp$.

To show $(\ker T)^\perp \subseteq \text{Im } T$, let $y \in (\ker T)^\perp$. Then

$$\|y - T(y)\|^2 = \langle y, \underbrace{y - T(y)}_{\in \ker T} \rangle - \langle T(y), y - T(y) \rangle$$

$$= -\langle y, T^*(y - T(y)) \rangle$$

$$= -\langle y, T(y) - T^2(y) \rangle = 0$$

Therefore $y = T(y) \in \ker(T - I) = \text{Im } T$, $(\ker T)^\perp = \text{Im } T$.

Now, $\ker T \subseteq (\ker T)^\perp = (\text{Im } T)^\perp$

Thm. Let V be a finite dim. inner prod space over F , and let $T: V \rightarrow V$ be a linear map.
Assume T has distinct eigenvalues $\lambda_1, \dots, \lambda_k \in F$.

Assume T is Normal if $F = \mathbb{C}$, T is Hermitian if $F = \mathbb{R}$.

For each $1 \leq i \leq k$, define

$$T_i: V \rightarrow V; \sum_{j=1}^n x_j \mapsto x_i \text{ where } x_j \in E_{\lambda_j}$$

Then, the followings hold

$$1) T_i \circ T_j = \delta_{ij} T_i, \text{ for any } 1 \leq i, j \leq k$$

$$2) I = T_1 + \dots + T_k$$

$$3) T = \lambda_1 T_1 + \dots + \lambda_k T_k.$$

Corollary.

If $F = \mathbb{C}$,

T is normal if and only if $g(T) = T^*$ for some $g(t) \in \mathbb{C}[x]$.

Proof.

$$g(t) = \sum_{i=1}^k \bar{\lambda}_i f_i(t) \text{ where } f_i(t) = \prod_{j \neq i}^k \frac{(t - \lambda_j)}{(\lambda_j - \lambda_i)}$$

Then, $g(\lambda_i) = \bar{\lambda}_i$ for all $1 \leq i \leq k$. Hence $g(t) = \sum_{i=0}^{k-1} a_i t^i$.

$$g(T) = a_{k-1} T^{k-1} + a_{k-2} T^{k-2} + \dots + a_1 T + a_0$$

$$= a_{k-1} (\lambda_1^{k-1} T_1 + \dots + \lambda_k^{k-1} T_k)$$

$$+ a_{k-2} (\lambda_1^{k-2} T_1 + \dots + \lambda_k^{k-2} T_k)$$

$\vdots$

$$+ a_1 (\lambda_1 T_1 + \dots + \lambda_k T_k)$$

$$+ a_0$$

$$= g(\lambda_1) T_1 + \dots + g(\lambda_k) T_k$$

$$= \bar{\lambda}_1 T_1 + \dots + \bar{\lambda}_k T_k$$

$$= \bar{\lambda}_1 T_1^* + \dots + \bar{\lambda}_k T_k^*$$

$$= T^*.$$

The converse is trivial.

Corollary, In $F = \mathbb{C}$,

T is unitary if and only if T is normal and $\chi(T)(\lambda) \Rightarrow \lambda = \pm 1$.

T is Hermitian if and only if T is normal and all eigenvalues are real.